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Does gender affect access to and utilization of rehabilitation services for persons with amputations in Southern Ethiopia?

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Abstract

Background This study aims to understand the perspectives of healthcare workers who are involved in the care of patients about the factors that impact access to and participation in rehabilitation services for people with amputation in Southern Ethiopia, with a particular focus on gender aspects of both persons with amputation and healthcare workers.

Methods A qualitative study was conducted using semi-structured one-to-one, face-to-face interviews with 22 eligible healthcare workers from hospitals and rehabilitation centers in Southern Ethiopia. This study employed a Community-Based Participatory Research (CBPR) approach from 14 October 2022 to 16 January 2023 using consultative workshops with experts and engaged community members as Local Research Advisory Committees (LRACs). Two research assistants conducted interviews, which were recorded, transcribed, and securely stored in a password-protected cloud. The data were then analyzed using reflective thematic analysis. The research team reviewed and agreed upon the themes, sub-themes, and their meanings, using mind-maps to highlight interconnectivity.

Results The study identified five themes with interconnecting sub-themes: Barriers to accessing rehabilitation services (knowledge and awareness of rehabilitation services, fear and violence, culture and belief, long distance, transportation challenges and financial constraints, physical environment and, acceptability and utilization of prosthesis and rehabilitation services related barriers), Resources (quality of materials and physical infrastructure, and workforce), Rehabilitation system (rehabilitation to support independent living, engagement, and service design), Perceptions of patient's experiences (psychological/emotional, relationships and copings, and physical), and Health outcomes (pain, wound, and prosthetic discomfort).

Conclusions This study provides a unique contribution to understanding access (barriers/ issues) to rehabilitation, particularly highlighting the complex issues faced by women with amputations. The insights of health workers illustrate the multifaceted struggles of women and the reality of compromising their rights to equal and personalized rehabilitation services. It highlights the need for gender-inclusive programs, addressing financial constraints, limited decision-making capacity, and transportation issues. Community-based programs, cultural expectations, and

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innovative prosthetic design approaches are crucial for empowering women as well as promoting gender diversity in healthcare professionals.

Keywords Rehabilitation service, Gender, Healthcare Workers, People with Amputation, Community Based Participatory Research (CBPR), Ethiopia

Background

Amputation has life-changing and profound consequences at physical, emotional, and economic levels [1–3]. The degree of disability and quality of life after amputation depend on factors like access to, quality and timing of rehabilitation services and assistive technology (AT). Societal attitudes and acceptance of disability within the affected person's community and family support play a crucial role in this process [4]. While amputation incidence has decreased over the years in high income countries [5], this is not the case for many low- and middle-income countries (LMICs). The same holds true for traumatic amputation [6]. Here, amputation prevalence is higher due to insufficient prevention, health promotion and health care programs, especially related to diabetes management, and lack of accident safety and security measures [7]. This is further exacerbated in countries affected by conflict, either directly, or indirectly by influx of displaced populations or a previous conflict history resulting in long-term landmine contamination and health system challenges [8, 9]. Also, in many LMICs staff and material resources to perform advanced reconstructive limb-saving reconstructive surgery is just not there, so if the limb injury is severe enough, amputation is the only option. The stigma, physical and attitudinal barriers that persons with amputation (PwA) face in many contexts reduce their opportunities to access education, health services and the job market, reinforcing the interlinked circle of disability and poverty [10]. A LMICs with high amputation incidence therefore present high demands for rehabilitation, yet face particular challenges related to low supply of rehabilitation workforce and services [11]. In addition to financial difficulties of PwAs, there are significant shortages related to rehabilitation infrastructure, AT material supply, training programs, and workforce. Amputation management and rehabilitation usually comprise several, well-defined stages and a multidisciplinary approach [12, 13]. Furthermore, adherence to WHO standards and recommendations is essential to ensure best practices [14, 15]. However, implementing a best practice protocol for multidisciplinary amputation management for all in need remains unfeasible in many contexts [8].

In Ethiopia, non-communicable diseases such as diabetes, stroke, and hypertension are major contributors to disability, in addition to factors such as road traffic accident (RTA). Among Ethiopian adults, over 3% are reported to have diabetes and over 18% hypertension [16, 17]. According to a 2016 WHO report, the road traffic accident fatality rate in Ethiopia was 26.7 deaths per 100,000 people [18]. The RTA survivors are likely to present lifelong disabilities such as amputation, adding to the rehabilitation needs of an aging population with an increasing non-communicable diseases prevalence.

Although there is no well-documented data, the available evidence suggests that there are currently only 15 physical rehabilitation centers in Ethiopia [19] for a population of 126.5 million, each facing numerous challenges [20]. In many LMICs including Ethiopia, the proportion of PwAs with access to timely quality prosthetic fitting and comprehensive rehabilitation are likely to not constitute a representative sample of all in need, with women and girls at a disadvantage [8, 21, 22]. Access is commonly defined by five dimensions: affordability, availability, accessibility, accommodation, and acceptability [23, 24]. It is well researched and documented that gender influences the experience of these dimensions [25–27].

Illness and resultant disability can impact men and women differently due to traditional gender norms that affect access to and participation in healthcare. In Malawi, women were less likely to secure financial aid for healthcare and more likely to underuse essential healthcare for themselves. Women and girls were less likely to be prioritized for medical treatment by both genders [25]. In a study of young adults in Malawi, more women accessed healthcare services than men, largely due to pregnancy or child health, while men's visits were related to acute conditions. There were significant time costs for women associated with healthcare visits, with median time at the clinic and travel time twice that of men [28].

Previous work has shown remarkably higher proportions of male rehabilitation users in conflict-affected or low-resource settings suggesting access barriers for women and girls that require further investigation [8, 21]. There is a lack of evidences on health care providers' perspectives regarding gender aspects of access to rehabilitation services for persons with amputation in Ethiopia and beyond. Therefore, the multi-country research GRACE Project ("Gender aspects in Rehabilitation of Person with Amputation in Conflict-affected Environments") was launched [29].

To explore how gender aspects impact access to and participation in rehabilitation, it is crucial to understand the perspectives of those delivering these services. The perceptions, beliefs and attitudes of healthcare workers towards women with disabilities may influence

rehabilitation service seeking behavior of women, the way women are treated and, therefore, rehabilitation outcomes. Exploring the multidimensional perspectives on access to rehabilitation services is beyond the scope of this paper. As this research is part of a larger study, we encourage readers to consult the findings of the ongoing study once it has been completed.

This study aims, therefore, to understand the perspectives of healthcare workers who involved in the care of persons with amputation about the factors that impact access to and participation in rehabilitation services for people with amputation in Southern Ethiopia, with a particular focus on gender aspects of both persons with amputation and healthcare workers.

Methods

Research design and approach

We used a phenomenological design, which is a systemic qualitative research that allow the researcher to explore the lived experience and provide a deep insight into individuals' 'stories', perspectives and understandings [30].

This study builds on Community Based Participatory Research (CBPR) approach, which emphasizes the importance of creating partnerships between researchers and local research beneficiaries ("knowledge users") [31] to ensure that participation and influence of non-academic partners is central to the process of creating and applying knowledge. This approach was chosen as it 'equitably involves all partners; [patients, community members, health workers and the researchers] in the research process and recognizes the unique strengths that each brings' [26]. It was crucial that non-academic partners ('community') were central to the project and that the research strategies fostered a balance of power between the researchers and the community, to ensure that the research and new knowledge generated was relevant and of value to the community and would contribute to sustainable solutions to address challenges in access and participation in rehabilitation for people with amputation. Here we describe the multistage approach (Fig. 1) employed in setting up the CBPR project in the initial phase.

The interdisciplinary international research team had been working together since 2020 in the broad area of rehabilitation in conflict affected and low resource contexts and co-designed the project, underpinned by CBPR, during eight rounds of online workshops and meetings every two weeks which aimed to define the 'community', understand the context (community strengths and dynamics), define the geographic boundaries and identify stakeholders and research partners in the context of Ethiopia. The core research team, local research lead,

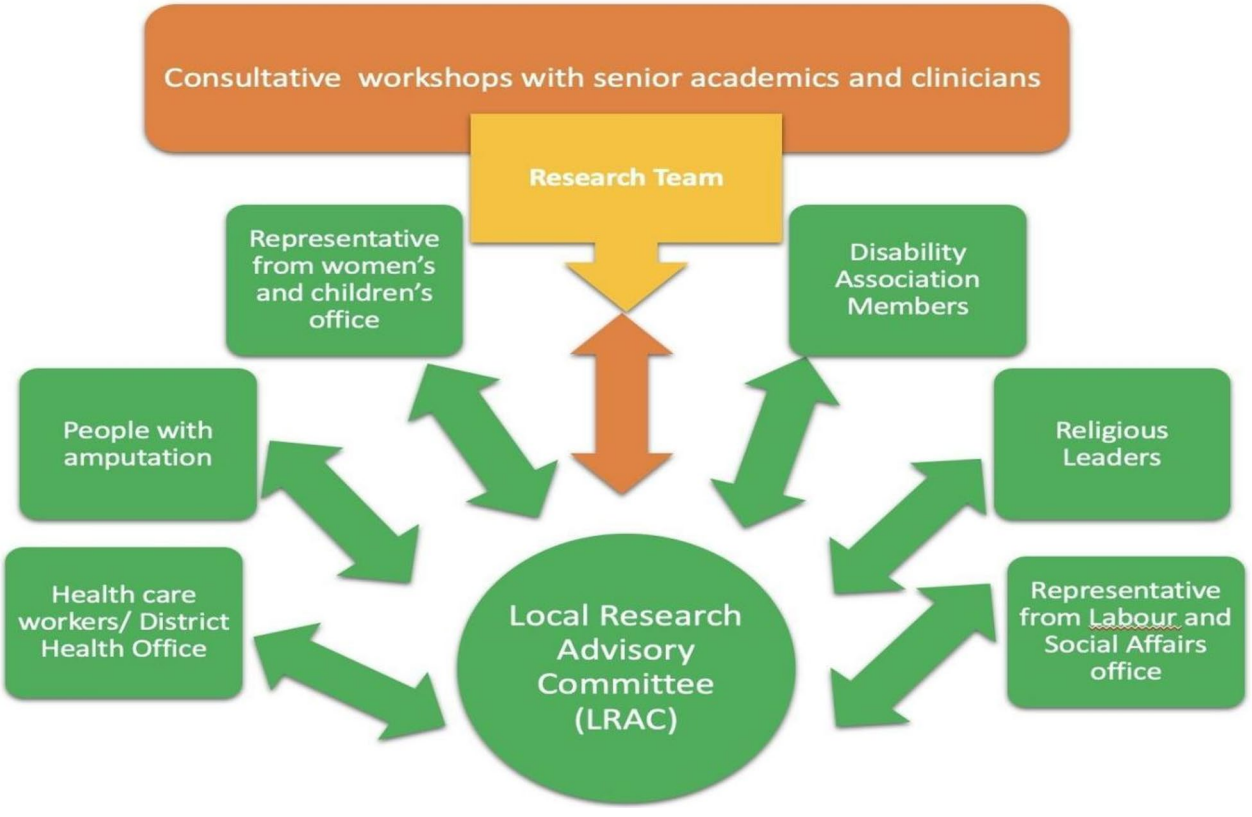


Fig. 1 Road-map for Community Based Participatory Research Approach in the initial phase

research assistants, and selected clinicians and researchers from study area participated in online workshops throughout the duration of the study project.

Partner engagement

In line with usual procedures in Ethiopia and to ensure that the project had permissions and support of high-level stakeholders, we first organized two rounds of hybrid consultative workshops with senior academics and clinicians from Wolaita Sodo University (WSU) and the affiliated comprehensive specialized hospitals who had an in-depth understanding of local contexts; to introduce the project, its purpose and the rationale for and principles of the CBPR approach. A key output from these workshops was identification of a diverse range of potential community partners/collaborators, reflective of key local stakeholders and a clear road-map of how the CBPR approach would be operationalised for this project.

Potential community members were identified and invited by the lead researcher (FE) via social and labor offices, and disabled people organizations and associations to participate as advisors throughout the study, recognizing their expertise in understanding the research context. To operationalize the CBPR approach, Local Research Advisory Committees (LRACs) were formed in each study setting (Arba Minch/ Wolaita Sodo, Fig. 1).

Members of the Local Research Advisory Committees were members of the research team (FE), people with amputation, health care workers, religious leaders, Disability Association members and representatives for the Women's and Children's Office and the Labour and Social Affairs Office. In each area, (Arba Minch, Wolaita Sodo) the Local Research Advisory Committee was split into a men's and a women's group to ensure truthful contributions, in respect of many, but in particular gender-related issues.

Initial Local Research Advisory Committees meetings explained the purpose of the GRACE project and clarified members' expectations of the process and in what way they could contribute. The LRAC members provided valuable insights, including lived experience, relevant to the study topic and research question and advised on participant recruitment. Though the interview guides were initially developed by the research team by reviewing relevant literature [1, 2, 4] in alignment with the study's objectives, the LRACs were also consulted. They reviewed and revised the interview guide for semi-structured interviews. The full interview guide is available in Supplementary 1. Each LRAC met three times within 12 months and provided ongoing support and guidance for successful completion of the study. The research team together with LRAC members reviewed the preliminary findings from the first 11 interviews and provided guidance on points to address during subsequent interviews. Final findings were shared with LRAC members during the final meeting.

Study participants and selection process

As part of a larger study, healthcare workers involved in the care of PwA were invited to participate in interviews for this study. Participants were identified via purposeful diversity sampling, ensuring a wide range of healthcare workers, see Table 2, across a large regional specialized hospital, a general hospital, well known for amputation surgeries and acute post-operative care, and a large regional rehabilitation center and prosthetic workshop. Both settings are part of the public health services in Ethiopia, financed by a combination of the Ministry of Health, community health insurance scheme and out-of-pocket patient expenditure.

The study was conducted from October 14, 2022, to January 16, 2023, in purposively selected settings namely Arba Minch and Wolaita Sodo towns in Southern Ethiopia; approximately 454 km and 330 km south of Addis Ababa, respectively. There are two general hospitals, one comprehensive specialized teaching hospital and one Rehabilitation center in the two study settings. The hospitals offer physiotherapy services alongside other routine medical care. Notably, the Rehabilitation Center, the only facility of its kind in the Southern Ethiopia region, provides both acute and chronic physical rehabilitation services. Located near the general hospital, the center facilitates care for patients requiring acute rehabilitation through a referral linkage system.

Data collection

Two research assistants were employed to conduct over 21 hours' interviews and training was provided by the Principal Investigator (COS, online) and FE. The training was undertaken over 2 sessions on 2 consecutive days. The first session consisted of an overview of the GRACE project, the CBPR approach and how this would be operationalised in the local context. The second session focused on training in relation to sampling and participant recruitment, interview schedule and technique (including role play), recording, transcription, translation and data management. FE observed the first two interviews of each research assistant. Both research assistants held a health-related post-graduate degree and had previous experience in conducting qualitative research and familiar with the study setting. Interviews were conducted in the local language (Amharic) and recorded using an audio recorder. Recordings were transcribed verbatim and translated into English by the respective interviewer. The lead researcher (FE) ensured translation accuracy by cross-checking the translated transcripts against the original audio recordings in the local language. All interview recordings and transcripts were

securely stored in a password protected cloud, accessible only to the research team.

Data analysis

Data were analyzed using reflective thematic analysis, guided by steps outlined by Braun and Clarke, with adherence to relevent theoretical assumptions [32, 33]. A constructionist epistemology in that 'meaningfulness' (rather than recurrence) was highly influential in the development and interpretation of codes and themes using Opencode software Version 4.0. This was strengthened by regular small team and wider GRACE team meetings and workshops to present codes and understand their meanings with respect to the Ethiopian context. Data analysis was led by FE, who has a deep understanding of the context, which was further strengthened by regular engagement and discussion with the LRACs. Experiential (rather than critical) orientation was adopted in our data analysis in order to emphasize meaning and meaningfulness as ascribed by participants. To provide context for the quotes presented in the study, a critical analysis was conducted, specifically examining any gender biases in the perspectives and experience of healthcare workers involved in the care of person with amputation. The study aim prioritized the lived experience of the study participants from different contexts and professional expertise. A predominantly inductive approach to analysis was adopted using open coding. A degree of deductive analysis was however undertaken to ensure that the open-coding contributed to producing themes that were meaningful to the research questions and the priorities raised by LRAC. The steps undertaken are described in Table 1 below. Initial data categories were organized into a codebook (step 3a below), reviewed (step 4 below) and agreed by the research team (step 5). Both semantic and latent coding were utilized.

Table 1 Steps undertaken during reflexive thematic analysis

1. Data familiarization	FE read and re-read all transcripts
2. Initial coding	FE reviewed all transcripts and generated initial codes, using open code software.
2a. Generation of data categories	FE presented initial codes to the research team during online meetings. Initial coding illustrated the breadth and complexity of the data. The study aim was reviewed and findings from the LRAC meetings were also presented. Large data categories were identified (from these sources) and agreed to by the research team. These categories and initial codes which underpinned them were organized into a codebook. This step was undertaken, not to follow a strict deductive approach, but rather to frame and facilitate further analysis of a large and complex data set.
2b. Generate codes	FE undertook a second round of line- by -line coding of all transcripts, and together with research assistants, reviewed codes for clarity and meaning
3.Generate themes	The codes were organized into categories (contained in the codebook), and new categories were added if necessary. These categories were then considered the first iteration of themes.
4. Review potential themes	Potential themes, arising from the data categories, were reviewed by the research team
5. Defining and naming themes	The themes, sub-themes and their meanings were reviewed and agreed by the research team during online meetings. To highlight the inter-connectivity between themes and sub-themes, mind-maps were created.
6. Report writing	

Trustworthiness

We applied Guba and Lincoln's criteria for credibility, transferability, dependability, and confirmability to ensure trustworthiness [34]. Credibility was achieved through member checks, with study participants and LRACs reviewing the findings, and consistent engagement with the data. The study's transferability was supported by purposeful sampling of study settings, participants, and LRAC members, detailed descriptions of participants while maintaining anonymity and the research process, and regular reflexive review of findings. Dependability was ensured by documenting the research process and meeting minutes, and confirmability was achieved by maintaining a reflective journal, documenting researchers' reflexivity and bracketing throughout the study. In addition, three independent researchers and clinicians, and LRACs reviewed the interview guide, and raw data to ensure reliability. Additionally, this study was conducted by a diverse and experienced team of researchers with strong reputations in qualitative research.

Results

Description of participants

A total of 22 interviewees (mean age 33 years, age range 24–52 years, 82% male, 54.5% working in hospital setting and work experience ranged from 1 to 2 months to 18 years), see Table 2, were included providing over 21 hours' of data.

We identified five interconnected themes and subthemes, all-encompassing gender dimensions, as depicted in Fig. 2 (Mind map) below: Barriers to Access rehabilitation services, Resources, Rehabilitation System, Perceptions of patient Experience, and Health Outcomes.

Barriers to access rehabilitation services

Knowledge and awareness of rehabilitation services Participants reported that PwA, regardless of gender, lacked knowledge and awareness of available rehabilitation services. Lack of interest by some PwA and social isolation was also deemed an obstacle to access services. Participants reported that gaps in the health system like absent or poor referral mechanisms and/or lack of

Table 2 Characteristics of study participants

Characteristics		Total, *N*
Age (Years)	Mean (SD) = 33 (6.33)	-
Sex	Male	18
	Female	4
Specialization	Orthopedic Surgeon	7
	Physiotherapist	7
	Nurse (Generic, Surgical, rehab)	3
	Orthopedic technician	3
	Social Work (CBR Worker)	1
	CBR Project Manager	1
Workplace	Hospital	12
	Rehabilitation Center	10
Work Experience (Years)	Median = 4 IQR = 10 (3–13)	-

CBR = Community based rehabilitation

continuity of care and remote living environments were reasons for the lack of information affecting access.

> *... There are people who don't have information. ... Some disabled people do not watch the media and are not aware of something. Some disabled people don't want to get out of their houses. The family also didn't listen to information about disability. ... I ask myself; are there people who didn't know our center while living here in Arba Minch Sikela (a village in Arba Minch Town, located a walking distance from the rehabilitation center). Do you believe this person (the one who lives near the center but didn't know where to get rehab services) is still alive? (Female, Orthopedic technician)*

Fear and violence Women with amputation were perceived to be at risk of experiencing violence and therefore, feared attending rehabilitation alone, while men with amputation encountered less threat.

> *Although a woman has no physical disability, she is vulnerable to physical violence. When she is disabled, she becomes more susceptible. A physically disabled woman could be exposed to violence when there is no one. She couldn't escape the attack by running away due to her disability. Easily suffer from any type of threat whether natural disaster, man-made assaults or sexual violence. (Male, Orthopedic technician)*

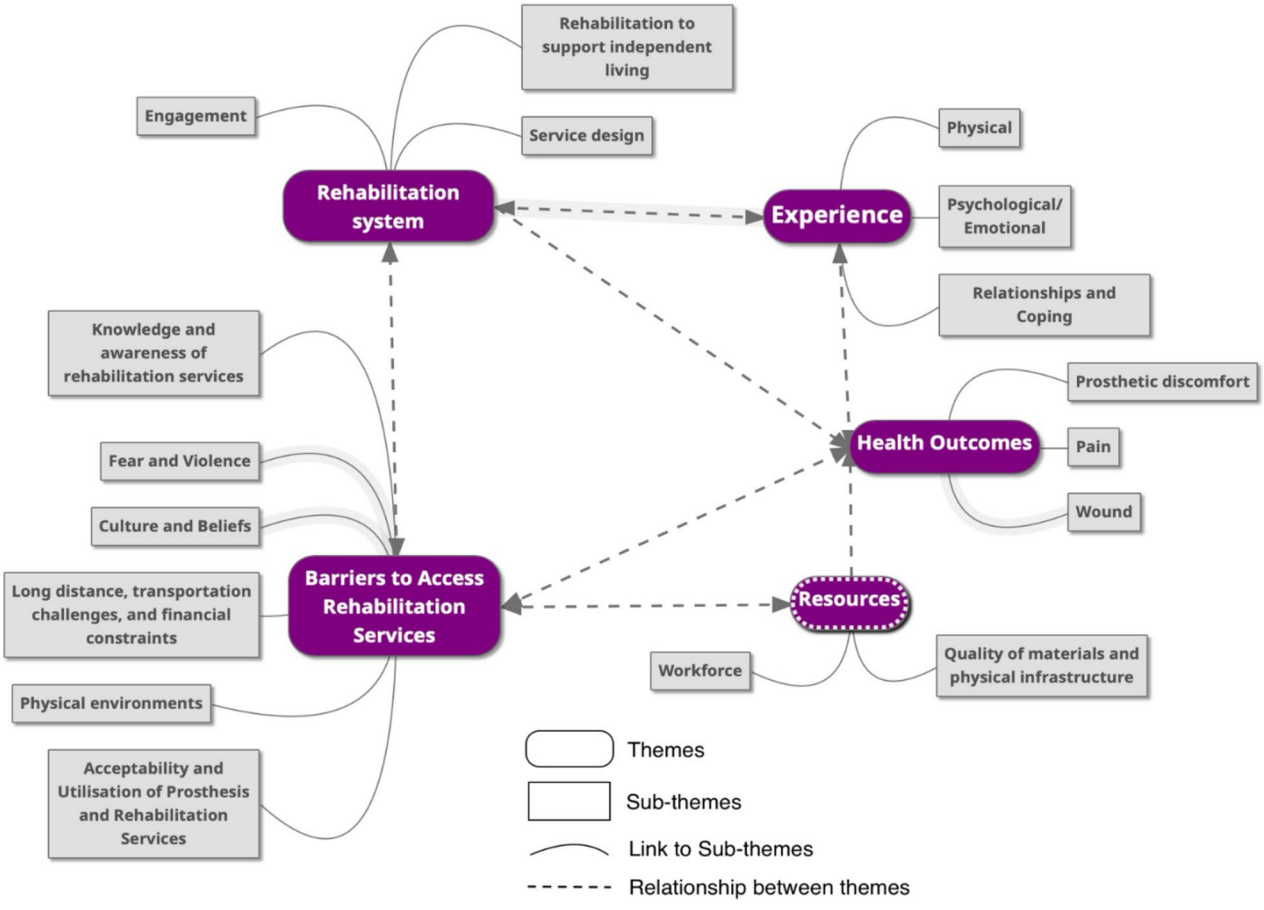


Fig. 2 Mind Map of interconnected themes and sub-themes

Risk of specific sexual and gender-based violence was disclosed as a barrier:

> *… The family by itself, if she is a woman, wouldn't send. They are suspicious of sexual violence. They fear that she may be sexually attacked. So, they say "Not this time, you will take the service in the future slowly, skip this time. (Male, Rehab Nurse)*

Culture and Beliefs Participants reported society's negative attitude towards women with amputation, stigmatizing them more than men with amputation.

> *…. We used to see a female (disabled) who was tied in the kitchen like a sheep… we asked why they did that up on her and they told us that "… as the guests see the disabled girl, they may say that "it is due to curse or something from genetics". Therefore, it is good for her to be tied in the kitchen rather than somebody insulting us …" her father said. He confidently said that he tied her. (Female, CBR worker)*

Local beliefs and attitudes dictate that women with amputation are busy with household chores and taking care of family.

> *"…women are undoubtedly affected. being a woman in and of herself, and … she is in charge of the family. Women look after the family more than men do (Male, Physiotherapist).*

Men with amputation were perceived to overcome challenges to access rehabilitation services since they were breadwinners.

> *"In our culture, men are engaged in so many activities because he is the head of household. It is the man that fulfills what is needed for his family. As a result, men overcome challenges because they are the leader of their family and have responsibility (Male, Physiotherapist).*

Long distance, transportation challenges, and financial constraints A consistent sub-theme throughout was poverty as a significant access barrier, even if rehabilitation services were free of charge, as financial capacity is required for transport and other costs associated with attending rehabilitation. This is amplified in cases where people live in rural, distant communities.

> *I don't think that there is another challenge except for the financial challenge. Some females depend on others, who depend on their families. Some families don't want this because of fear of finances. …. Some people used to say "We don't want him/her treated, we will go home", such people may say that "We don't have money to go for rehabilitation services and transportation. (Female, Surgical Nurse)*

The participants disclosed that although the rehabilitation services are free of charge for the poor, it is more challenging and expensive for women with amputation to have it, especially those who lack education, insurance and economic means. One of the participants said:

> *It is expensive to get it. If they didn't get here in Arba Minch, they may go to Addis Ababa or Assela. If you go to Addis Ababa, calculate your cost. Some services require payments; …. It is not free of charge. So, for women, regarding the economy, it is difficult. (Male, orthopedic technician)*

Economic challenges had increased with recent conflicts, disproportionately affecting rural populations, socioeconomically deprived groups, and women with amputation.

> *"…. where do they get such a huge amount of money? Some people have an interest in using the service but stay at home due to financial challenges. (Female, CBR Worker)*

Physical environments Participants expressed that the residential environment, infrastructures, buildings, and topography were not accessible for PwA and hindered access to rehabilitation services. Participants further described that the physical environment matters even after accessing the rehabilitation service for PwA, particularly for women with amputation and those from rural areas.

> *If there is a long distance and if she is a woman, she has to cross different places. She may think of many things. … There are women who say "How could I go carrying this?". … A person (women with amputation) living in mountains, muddy and highland, if you say use the artificial [prosthesis] and go, it may be difficult. It may stick with mud and may be slippery. In hot areas, it may create warmth. There is a tendency of throwing it off. So, the geographical location and weather condition of their living area matters. (Male, Physiotherapist)*

Acceptability and Utilisation of Prosthesis and Rehabilitation Services Participants described that women with amputation do not want to be served by male providers, need cosmetically sound prosthesis and some had a fear of using the prosthesis.

> *Women by nature love cosmetics. While using artificial [prosthesis], she may want to do what her friends did…. For one physically disabled man as*

long as he gaits and walks, he doesn't worry even if the artificial [prosthesis] is cosmetically poor. Women, beyond using them for walking, need the artificial [prosthesis] to be attractive to the people. They prefer the artificial not to be seen. (Male, Orthopedic technician)

Participants also reported instances where PwA did not accept the rehabilitation service due to their negligence, they either preferred to live with begging or found that the available rehabilitation services didn't meet their needs.

…using prosthesis may bring them to normality but they want to live with begging, there are many people who say "I don't want to use" … They (PwA) think people may stop caring, they take crawling on the ground as an alternative source of income and not quit begging. (Male, Project manager)

Resources

Quality of materials and physical infrastructure Shortage of mobility-related assistive devices, consumables, and appliances were barriers commonly expressed by participants. Despite the shortage, the healthcare workers used some alternative consumables (e.g. socks) and counseled the PwA to use them sparingly.

… this rehabilitation center is most of the time unfunctional due to lack of materials. If (name of donor) interrupted the delivery of these raw materials, there would be nothing. …it is public but there is no way we can get raw materials. …To fulfill women's needs, we can't attain their needs with the current situation. Because we can't buy from the market what she prefers.… The demand is high and the supply is very small. (Male, Orthopedic technician)

Despite advancement in prosthetic technology globally, there are shortages and currently available models do not seem to meet the needs of women with amputation. The participants reported that women with amputation frequently complained that the prosthesis available in the center didn't meet their cosmetic needs.

…most of the time women not only see functionality but also, they need a cosmetically sound prosthesis. (Male, Orthopedic technician)

Besides cosmetic value participants expressed the impact of the weight of the prosthesis:

… if we come to marriage, when they become pregnant it is difficult for them to move using artificial means. … It may have weight and have more effect on the women. So, artificial is more suitable for men. (Male, Physiotherapist)

The devices were generally not comfortable, old and poor quality models that didn't meet the needs of women with amputation as noted by participants.

Our prosthesis is old. It was given even before I joined this hospital. It is not up to date. We are still using the same prosthesis. … It is too heavy for (women). (Male, Physiotherapist)

Participants further described room shortage as a challenge to do an assessment and to provide gender-specific dormitory services in separate rooms for PwA.

If their legs get amputated from their hips, there should be a separate room/place for men and women because women are ashamed when changing their clothes… (Male, Physiotherapist)

Workforce Participants noted the challenges with professional career pathways, staff shortage, and significant gender differences. The critical shortage of female professionals was frequently reported and also perceived as a barrier to attending rehabilitation.

They (female PwA) also challenge us for measurement. They say "There must be a female professional to treat us". Most of the health professionals here are men. I am the only female professional. They say "We (women with amputation) need only a female professional". They (women with amputation) fear. (Female, Orthopedic technician)

However, a male participant noted that women with amputation had limited access to rehabilitation centers and there was no choice to look for options regarding the gender of providers:

… you don't have an option about choice, you just use what is already available … You think about options when there is availability or an option to choose. I feel that they (women with amputation) prefer male professionals. (Male, Orthopedic Surgeon)

Participants noted that there were no psychology or psychiatry professionals at the rehabilitation center to provide psychological support. There was a need to recruit those professionals to provide standard service.

"Some people give up even if they do have money and can work by moving. … There is no psychiatrist

here. …. No one gives psychological treatment. It is very crucial. (Male, Orthopedic technician)

Rehabilitation System

Rehabilitation to support independent living Participants described different mechanisms needed within the rehabilitation systems to support PwA to lead an independent life. The need for different support mechanisms were noted such as psychological support, economic empowerment, and gender-sensitive rehabilitation.

So, it is better if first psychologically treated and then we physically rehabilitate and when they return to the community, it would be better if there is a mechanism that shows how to improve themselves financially. This is both for men and women. It is more difficult for women. (Male, Orthopedic technician)

Participants highlighted the need for task shifting, particularly for women with amputation, to address the financial challenges.

Better to bring (Women with amputation) to suitable areas like education, to a white collar. Better if they get closer and turn from labor work to mental work. In this way, we can solve the challenges they face economically. (Male, Physiotherapist)

Women with amputation were perceived to need more support and special attention as reported by participants.

"Regarding women, they need much support as compared to males. It is obvious. Especially if she is unmarried. It is very difficult. The attitude that exists in society is challenging. They need more support." (Male, Orthopedic surgeon).

Engagement Participants suggested the need for engaging PwA with health extension workers (paid community health workers), as well as rehabilitation professionals. Participants further described the need for engaging husbands in rehabilitation process for married or cohabiting couples.

…Sometimes despised by their husbands. Therefore, they should be encouraged, and let their husbands actively engage, their husbands should also be part of the solution and involve them starting from the decision making. (Male, Orthopedic surgeon)

Service design The current rehabilitation service needs to be re-designed to meet the needs of both men and women with amputation. As it has been expressed under different themes the existing rehabilitation services are designed for men with amputation as a norm and failed to meet mostly the needs of women with amputation.

The demand is high and the supply is very small. We can't fulfill women's interests. Their question is correct, asking for their matching material is acceptable. But we can't fulfill it in the current status. (Male, Orthopedic technician)

Participants expressed that the existing referral system is poor and there is no means of follow-up about PwA accessing the service. There needs to be a feedback system as suggested by participants.

It is difficult to know how the patients live, what challenges they face, their feelings, and existing gaps because we don't have contact with the patient once we refer them to the center, … We don't know whether the patients went to the referral center for prosthesis or not. (Male, Orthopedic surgeon)

Despite this reality, participants also noted that highlighting achievements of PwA as role models, were found to be effective.

…there are amputated people who lost their legs … and use artificial… We show others that he is productive, he lost his leg but he can move. We tell them that he is by far more affected than your case but he can walk (Male, Orthopedic surgeon)

Patient's experiences

Psychological/Emotional PwA were perceived to have varied psychological/emotional experiences as noted by participants going from depression, shock, shame to emotional confusion and the wish to die. Most of the psychological/emotional experiences were related to initial reactions to amputation and reportedly greater among women.

"Disability is more challenging for women. They would try their best to avert amputation decisions. They consider undertaking a suicide…. She says "Never for an amputation decision" (Female, Orthopedic technician).

Besides initial reactions, participants suggested that women with amputation face more psychological trauma as noted below.

If the patient is a man, he may have the potential to cope with societal challenges. It is more immense for women, there may be a feeling of losing fortune. So, most of the time women will be psychologically affected. (Male, Orthopedic surgeon)

Relationships and Coping Participants expressed that PwA experienced enormous relationship and coping related impacts. Participants noted that marriage-related challenges, discrimination, less chance to participate in social affairs, being economically dependent and a burden for the family impacted all, but particularly women with amputation.

> *… the chance to get married is less for female PwA.… Males have a chance to get married but if a female has mobility-related problems most males don't want to marry females of such kind. …There are still females who still haven't married. (Female, CBR Worker)*

However, participants believed that PwA could do anything and lead a better life. Amputation is not the end of life and there were brilliant and exemplary PwA who regained hope and were busy with their work.

> *…It is possible to marry an amputated girl and live with her and it is possible to live with an amputated man so people should know that it is not the end of the world." (Male, Orthopedic surgeon)*

Men with amputations were better at coping since their wives supported them. One participant said:

> *…. When it happens to men, the probability of them coping is high because their wives support them but I don't see much in turn among men when their wives get amputated. (Male, Orthopedic surgeon)*

Physical Informants reported that physical impairment associated with amputation can hinder participation. The lifestyle, area of residence, and body image further affected their ability to take part in activities that demand physical stamina. Participants further noted that women with amputation and those who reside in rural areas face more challenges.

> *… But, it affects women more because the man gives negligible attention to body parts; … If a black spot is noticed in my fingernail, I may not be concerned. But my wife may be very much concerned. That means they give great value to this thing (body image)… (Male, Orthopedic surgeon)*

One participant further noted that men with amputation from rural areas face social discrimination due to their physical disability.

> *… in the rural areas, there is what we call "DEBO" where people build someone's house together.… People don't invite them (men with amputation) because they say that "a PwA is incapable of doing these activities. How is he going to stand? (Female, CBR Worker)*

Health outcomes reported

Pain Participants described that PwA experienced phantom pain and pain related to wound and bone prominence in the residual limb area. Some participants perceived that women with amputation have lower pain thresholds. One participant said.

> *"Women's pain threshold or tolerance level is less but men are better at tolerating the pain. (Male, Orthopedic surgeon)*

Wound Participants described that delayed wound healing of the residual limb impacts patients' quality of life. Participants noted that unfitting prostheses, often gendered as highlighted earlier, were commonly reported by women with amputation and resulted in delayed wound healing. This, in turn, hindered timely prosthesis fitting and increased the length of stay at rehabilitation center. Additionally, women were less likely to seek care timely due to various barriers mentioned in other themes of this study, further increasing their risk of complications related to wound healing. *"…. It created a wound when using an artificial leg and came to us saying "change to me". I said to her before replacing this artificial (prostheses), the wound should heal with wound care. … She said "I wouldn't go without the artificial". … She stayed for more than a month." (Male, Orthopedic technician).*

Prosthetic discomfort Participants described that PwA complained about prosthetic discomfort. Participants further noted that women and men with amputation had different outcomes. Although women with amputation get familiarized and trained faster; prostheses fit men better.

> *"… (women with amputation) had problems such as the prosthesis being tight and not comfortable, and the amputated limb becoming weaker." (Male, Physiotherapist).*

Discussion

This study aims to understand the perspectives of healthcare workers who involved in the care of the patients about the factors that impact access to and participation in rehabilitation services for people with amputation in Southern Ethiopia, with a particular focus on gender aspects.

Understanding the perspectives of Ethiopian healthcare workers using a community-based participatory research approach demonstrated the complexity of gender issues for persons, particularly women, with amputations and their rehabilitation journey. The insights of healthcare

workers illustrate the multifaceted struggles of women with amputation and a reality which compromises their rights to equal and personalized rehabilitation services. These understandings are key to the development and provision of rehabilitation services.

To date, no study has addressed gender perspectives for access to rehabilitation services in Ethiopia and only a few have done so on access to healthcare services among women [35–37]. Our study highlighted a number of barriers to rehabilitation for people with amputation. As services tend to be concentrated in urban settings [22], travel to services has a significant impact on access and cost for persons living in remote and rural regions, and this is especially true for women with amputation seeking rehabilitation. Limited access to financial resources in households, safety concerns, and family commitments can make it difficult for women to leave home to receive services. Our findings echo studies of gender and access to healthcare service in Ethiopia [35–37] and lessons learned from other LMICs by the ReLAB-HS project [38] who reported that access to rehabilitation is challenging for women with disability when there is no appropriate transportation, companion or assistant for travel and while attending rehabilitation, equipment's and, accommodation in the rehabilitation sites.

Our findings highlighted that the residential environment has a double impact on PwA. It prevents PwA from accessing rehabilitation services and once they access, the topography is not suitable to use mobility aids. Our findings corroborate the study finding from southern Ethiopia that the living environment impedes their free movement [39] and doesn't meet the needs and interests of disabled people [40].

Our findings suggest that negative cultural traditions may impede access to rehabilitation for women. Women play a vital role in society, often caring for multiple generations [41, 42]and in times of crisis, often undertake extra tasks and roles to maintain social, education and healthcare systems and activities [43]. Enabling them to access rehabilitation and reach their potential has far-reaching societal impacts.

Participating centers suffered from staff shortages, particularly female staff shortage, inadequate supplies and appropriate equipment, similar to LMIC study findings [44], severely impacting the quality of care delivered.

A shortage of appropriately trained health workers is a significant barrier to development and delivery of a comprehensive rehabilitation programme. Available data (2023) indicate that there are less than 700 physiotherapy and rehabilitation professionals in Ethiopia, while the national Human Resources for Health Strategic plan aims to increase to 1300 by 2025. Lack of health workers is a key issue in Ethiopia [45].

Furthermore, some patients were reported to prefer to be treated by professionals of the same gender. Our findings were similar to a gender perspective study on physiotherapy treatment in patients with neck and back pain has some indication that patients prefer physiotherapists of the same sex/gender [46]. This highlights the importance of gender diversity in training physiotherapists in Ethiopia. Although the world physiotherapy data (2023) [45] show a nearly equal proportion of female physiotherapist professionals in Ethiopia, this does not reflect the situation in the study setting. This could be because the majority of physiotherapy professionals have changed careers due to less opportunity to upgrade their career, perceive the rehabilitation work is dangerous and laborious for women or prefer working in urban areas but not willing to work in rural settings [47, 48].

Similar to other studies [49, 50], the available prostheses failed to meet the needs of women with amputation. Participants reported that women expressed dissatisfaction with the design, weight, and overall appearance of their artificial limb.

The need for a comprehensive rehabilitation system, encompassing economic empowerment and psychological support, as well as physical rehabilitation, was evident [51]. Women [with disabilities] usually have less decision-making power and economic potential, and face stigma and discrimination when they return to the community [52]. As outlined in this study, begging has been considered an alternative or better source of income and solution for a crisis [53].

Studies have demonstrated increased levels of anxiety and depression [54] among people with amputation and the need for mental health interventions [55], as echoed in this study. There is strong evidence for mental health interventions [56] and economic initiatives such as providing education, employment and financial resource opportunities to improve their participation in development activities and alleviate poverty for people with disabilities [57].

Participants perceived that men with amputation cope better and return to routine activities after having mobility-related assistive devices. Men with amputation get support from their partners/wives that help them regain independence. In contrast, women with amputation have less functional gain, probably because the prostheses do not meet their needs as women [49] and they receive less care from their male partners [58]. Findings suggested that the devices were generally not comfortable, old and poor quality models that didn't meet the needs of women with amputation. This is in line with what has been reported in the Haiti study, they may be given lower quality or even used prostheses [50]. Similar to other studies, findings suggested that healthcare workers perceived gender differences in expression of pain/ pain

experiences, which may impact on how well women's pain is managed post amputation and during rehabilitation, with devastating consequences on rehabilitation., It is imperative that health workers and service managers receive training to ensure equitable quality of care for both women and men.

Strengths, limitations and future research

Our study addresses a dearth of research on this issue in and beyond Ethiopia. The available research on gender aspects in relation to rehabilitation services in LMICs is extremely limited. Furthermore, our employment of community-based participatory research proved effective in designing our approach, built on community input through the LRACs and gaining valuable insights from local health professionals. We engaged key community members at each step of this study. This study was completely guided by the LRACs and selected local researchers and expertise. We also trained local research assistants and built the capacity of local researchers including the lead researcher of this study. We also brought together different stakeholders and expertise in one table and ensured equity. We also committed ourselves for long term commitments by continuously engaging in this research area to address the key barriers identified.

The limitations of our study include that we sought perspectives from healthcare workers (HCWs) who closely interact with men and women with amputations. However, HCWs' insights may not reflect those of affected persons. We acknowledge that their contributions are two-sided: as much as they reveal the participants' views of women and men with amputations, they are also an illustration of their own attitudes and beliefs about this population, which can be steeped in personal biases, unconsciously contributing to access inequalities. Another remarkable constraint is the limited number of female professionals interviewed ($n = 4$). Most statements reflect thus the perspectives of male HCWs on men and women with amputation. A higher number of female participants would have likely shown different perspectives. We extended our criteria to include nurses and CBR professionals who work with PwA but didn't succeed in recruiting more than four female HCWs. Enhancing female representation in this field should be a focus of future research initiatives. This study was conducted in two specific settings, which may limit the direct transferability of the findings to other contexts that differ significantly from these locations. However, by providing thick descriptions and rich contextual data, we have aimed to enhance the potential for transferability to settings with similar characteristics.

Conclusions

This study highlighted a number of access barriers and issues, notably for women with amputation. The health workers' perspectives illustrate the multifaceted struggles of women and the reality of compromising their rights to equal and personalised services in hospital and physical rehabilitation centers. It highlights the need for action at the policy level, which includes advocacy for gender-inclusive rehabilitation services and acknowledging the specific challenges faced by women with amputation.

Policy changes should address poor referral linkage between hospitals and rehabilitation centers, financial constraints, limited decision-making capacity, and transportation issues that disproportionately affect women. We encourage the development of community-based rehabilitation programmes that empower women with amputation economically, such as vocational training and income-generating activities, and that meet their unique needs, considering the impact of cultural expectations.

We call for a significant and systematic investment and increase in the workforce involved in rehabilitation services, with a particular focus on gender diversity. Initiatives for capacity building and career development should specifically target female professionals in this field. Further research is warranted on how the gender of healthcare professionals affects the utilization of rehabilitation services, with a focus on the voices of persons with amputation. Health workers must be trained to ensure equal care for women and men. Innovative approaches are required to enhance prosthetic design for women, considering factors such as weight, appearance, and functionality. Future studies should explore the correlation between prosthetic design, pain, rehabilitation interventions, overall satisfaction, and long-term outcomes and expected functional requirements for both genders.

On a community level, it is advised to implement community awareness programs to dispel stigma associated with amputation and rehabilitation. Promoting the importance of rehabilitation by emphasizing the societal benefits is key, especially for women. Within communities. We advise comprehensive assessments of rehabilitation systems, ensuring they are responsive to the diverse needs of both men and women with amputation. Finally, multi-level, multi-stakeholder collaborations and continued community engagement are warranted to sustain engagement and ensure ongoing input and innovation.

These recommendations aim to foster a holistic and inclusive approach to rehabilitation services, addressing the unique challenges faced by women with amputation and advancing gender equality in healthcare.

Abbreviations

and Acronym

AT Assistive Technology

CBPR	Community based participatory research
ERC	Ethical Review Committee
GRACE	Gender aspects in Rehabilitation service among amputations in conflict affected environments
HCWs	Healthcare Workers
LRACs	Local Research Advisory Committees
LMICs	Low- and middle-income countries
PwA	Persons with amputation
RTAs	Road traffic accidents
UCD	University College Dublin and
WSU	Wolaita Sodo University

Supplementary Information

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Supplementary Material 1

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Author contributions
FE, COS, and CAB: Conceptualization; Data curation; Formal analysis; Funding acquisition; Investigation; Methodology; Project administration; Resources; Supervision; Validation; Visualization; Roles/Writing - original draft; and Writing - review & editing. AW: Conceptualization; Funding acquisition; Methodology; Project administration; Resources; Visualization; and Roles/Writing - review & editing. AHE: Methodology; Supervision; Validation; Visualization; Roles/Writing - review & editing. All authors read and approved the final manuscript.

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Data availability
The anonymized datasets that support the findings of this study are available from the corresponding author upon reasonable request.

Ethics Declarations

Ethics approval and consent to participate
Ethical approval was obtained from the Ethical Review Committee (ERC) of College of Health Science and Medicine, Wolaita Sodo University, Ethiopia *[Ref No CRCSD 1/6/02/14]*. All potential participants were informed about the study via a study information leaflet and invited to participate. All interested participants were included and signed a consent form prior to the interview.

Competing interests
The authors declare no competing interests.

Consent for publication
Not applicable.

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References

1. Darter BJ, Hawley CE, Armstrong AJ, Avellone L, Wehman P. Factors Influencing Functional Outcomes and Return-to-Work After Amputation: A Review of the Literature. J Occup Rehabil. 2018;28:656–65. https://doi.org/10.1007/s10926-018-9757-y.
2. Horgan O, MacLachlan M. Psychosocial adjustment to lower-limb amputation: A review. Disabil Rehabil. 2004;26:837–50. https://doi.org/10.1080/09638280410001708869.
3. Maciver M, Dixon D, Powell D. Quality of life in young people with limb loss: a systematic review. Disabil Rehabil 2023:1–12. https://doi.org/10.1080/09638288.2023.2270908
4. Sayeed MSI, Oakman J, Dillon MP, Stuckey R. Influential factors for access to and participation in rehabilitation for people with lower limb amputation in East, South, and Southeast Asian developing countries: a scoping review. Disabil Rehabil. 2022;44:8094–109. https://doi.org/10.1080/09638288.2021.1994025.
5. Moxey PW, Gogalniceanu P, Hinchliffe RJ, Loftus IM, Jones KJ, Thompson MM, et al. Lower extremity amputations - a review of global variability in incidence. Diabet Med. 2011;28:1144–53. https://doi.org/10.1111/j.1464-5491.2011.03279.x.
6. Yuan B, Hu D, Gu S, Xiao S, Song F. The global burden of traumatic amputation in 204 countries and territories. Front Public Health. 2023;11:1258853. https://doi.org/10.3389/fpubh.2023.1258853.
7. El-Kebbi IM, Bidikian NH, Hneiny L, Nasrallah MP. Epidemiology of type 2 diabetes in the Middle East and North Africa: Challenges and call for action. World J Diabetes. 2021;12:1401–25. https://doi.org/10.4239/wjd.v12.i9.1401.
8. Barth CA, Wladis A, Blake C, Bhandarkar P, Aebischer Perone S, O'Sullivan C. Retrospective observational study of characteristics of persons with amputations accessing International Committee of the Red Cross (ICRC) rehabilitation centers in five conflict and postconflict countries. BMJ Open. 2021;11:e049533. https://doi.org/10.1136/bmjopen-2021-049533.
9. Oppong JR, Kalipeni E. The Geography of Landmines and Implications for Health and Disease in Africa: A Political Ecology Approach. Afr Today. 2005;52:2–25. https://doi.org/10.2979/AFT.2005.52.1.2.
10. Takasaki Y. Impacts of disability on poverty: Quasi-experimental evidence from landmine amputees in Cambodia. J Econ Behav Organ. 2020;180:85–107. https://doi.org/10.1016/j.jebo.2020.09.027.
11. Kamenov K, Mills J-A, Chatterji S, Cieza A. Needs and unmet needs for rehabilitation services: a scoping review. Disabil Rehabil. 2019;41:1227–37. https://doi.org/10.1080/09638288.2017.1422036.
12. World Health Organization. Package of interventions for rehabilitation. Module 2. Musculoskeletal conditions. Geneva: World Health Organization; 2023.
13. International Committee of the Red Cross. Prosthetic Gait Analysis for Physiotherapists. Geneva: International Committee of the Red Cross; 2014.
14. World Health Organization. WHO standards for prosthetics and orthotics. Geneva, World Health Organization. : 2017. License: CC BY-NC-SA 3.0 IGO. https://iris.who.int/bitstream/handle/10665/259209/9789241512480-part2-eng.pdf?sequence=2 (accessed January 31, 2025).
15. World Health Organization, United Nations Children's Fund (UNICEF). Global report on assistive technology. Geneva: World Health, Organization. and UNICEF; 2022. License: CC BY-NC-SA 3.0 IGO. https://iris.who.int/bitstream/handle/10665/354357/9789240049451-eng.pdf?sequence=1 (accessed January 31, 2025).
16. The International Diabetes Federation (IDF). Diabetes in Ethiopia. (2021). The International Diabetes Federation (IDF) 2021. https://idf.org/our-network/regions-and-members/africa/members/ethiopia/ (accessed December 12, 2023).
17. Norris T, Girma M, Genye T, Hussen A, Pradeilles R, Bekele Z, et al. Predictors of prediabetes/diabetes and hypertension in Ethiopia: Reanalysis of the 2015 NCD STEPS survey using causal path diagrams. Matern Child Nutr. 2022. https://doi.org/10.1111/mcn.13365.

18. Global Road Safety Facility. Ethiopia's Road Safety Country Profile. Global Road Safety Facility 2016. https://www.roadsafetyfacility.org/country/ethiopia (accessed December 12, 2023).
19. UNPF. World Population Dashboard Ethiopia. UNPF 2023. https://www.unfpa.org/data/world-population/ET (accessed December 12, 2023).
20. MOH Ethiopia. National Rehabilitation and Assistive Technology Services Management Guideline. WHO 2020. https://cdn.who.int/media/docs/default-source/assistive-technology-2/ethiopia-national-rehabilitation-and-at-services-management-guideline.pdf?sfvrsn=22fc4dc8_9 (accessed December 12, 2023).
21. Barth CA, Wladis A, Blake C, Bhandarkar P, O'Sullivan C. Users of rehabilitation services in 14 countries and territories affected by conflict, 1988–2018. Bull World Health Organ. 2020;98:599–614. https://doi.org/10.2471/BLT.19.249060.
22. Matin BK, Williamson HJ, Karyani AK, Rezaei S, Soofi M, Soltani S. Barriers in access to healthcare for women with disabilities: a systematic review in qualitative studies. BMC Womens Health. 2021;21:44. https://doi.org/10.1186/s12905-021-01189-5.
23. Obrist B, Iteba N, Lengeler C, Makemba A, Mshana C, Nathan R, et al. Access to Health Care in Contexts of Livelihood Insecurity: A Framework for Analysis and Action. PLoS Med. 2007;4:e308. https://doi.org/10.1371/journal.pmed.0040308.
24. Penchansky R, Thomas JW. The Concept of Access. Med Care. 1981;19:127–40. https://doi.org/10.1097/00005650-198102000-00001.
25. Azad AD, Charles AG, Ding Q, Trickey AW, Wren SM. The gender gap and healthcare: associations between gender roles and factors affecting healthcare access in Central Malawi, June–August 2017. Archives Public Health. 2020;78:119. https://doi.org/10.1186/s13690-020-00497-w.
26. Mwachofi AK. Gender difference in access and intervention outcomes: the case for women with disabilities. Disabil Rehabil. 2009;31:693–700. https://doi.org/10.1080/09638280802305929.
27. Sen G, Östlin P. Gender inequity in health: why it exists and how we can change it. Glob Public Health. 2008;3:1–12. https://doi.org/10.1080/17441690801900795.
28. Yeatman S, Chamberlin S, Dovel K. Women's (health) work: A population-based, cross-sectional study of gender differences in time spent seeking health care in Malawi. PLoS ONE. 2018;13:e0209586. https://doi.org/10.1371/journal.pone.0209586.
29. The Center for Disaster Medicine and Traumatology LU. GRACE. Gender aspects in Rehabilitation of persons with Amputation in Challenging Environments 2023. https://www.katastrofmedicin.se/rehabilitation-after-amputation (accessed November 10, 2023).
30. Groenewald TA, Phenomenological Research Design Illustrated. Int J Qual Methods. 2004;3:42–55. https://doi.org/10.1177/160940690400300104.
31. Jull J, Giles A, Graham ID. Community-based participatory research and integrated knowledge translation: advancing the co-creation of knowledge. Implement Sci. 2017;12:150. https://doi.org/10.1186/s13012-017-0696-3.
32. Braun V, Clarke V. Using thematic analysis in psychology. Qual Res Psychol. 2006;3:77–101. https://doi.org/10.1191/1478088706qp063oa.
33. Braun V, Clarke V. Conceptual and design thinking for thematic analysis. Qualitative Psychol. 2022;9:3–26. https://doi.org/10.1037/qup0000196.
34. Guba EG. Criteria for assessing the trustworthiness of naturalistic inquiries. ECTJ. 1981;29:75–91. https://doi.org/10.1007/BF02766777.
35. Demsash AW, Walle AD. Women's health service access and associated factors in Ethiopia: application of geographical information system and multilevel analysis. BMJ Health Care Inf 2023;30. https://doi.org/10.1136/bmjhci-2022-100720
36. Bunare A, Weldegebreal F, Habte S, Ketema I, Worku T. Utilization of Rehabilitation Services and Associated Factors Among Adults with Cancer Diagnoses at Hawassa Comprehensive Specialized Hospital, Ethiopia. Patient Prefer Adherence 2022;Volume 16:1571–80. https://doi.org/10.2147/PPA.S370923
37. Wharton-Smith A, Rassi C, Batisso E, Ortu G, King R, Endriyas M, et al. Gender-related factors affecting health seeking for neglected tropical diseases: findings from a qualitative study in Ethiopia. PLoS Negl Trop Dis. 2019;13:e0007840. https://doi.org/10.1371/journal.pntd.0007840.
38. Ott. & Morgan. Why sex and gender matter in interventions and research for rehabilitation within health systems. Learning, Acting and Building for Rehabilitation in Health Systems Consortium (ReLAB-HS) 2021.
39. Sona BD. Psychosocial Challenges of Women with Disabilities in Some Selected Districts of Gedeo Zone, Southern Ethiopia. Int J Criminal Justice Sci (IJCJS). 2015;10:173–86.
40. Ayalew AT, Adane DT, Obolla SS, Ludago TB, Sona BD, Biewer G. From Community-Based Rehabilitation (CBR) Services to Inclusive Development. A Study on Practice, Challenges, and Future Prospects of CBR in Gedeo Zone (Southern Ethiopia). Front Educ (Lausanne). 2020;5. https://doi.org/10.3389/feduc.2020.506050.
41. Barigozzi F, Cremer H, Roeder K. Caregivers in the family: Daughters, sons and social norms. Eur Econ Rev. 2020;130:103589. https://doi.org/10.1016/j.euroecorev.2020.103589.
42. Bracke P, Christiaens W, Wauterickx N. The Pivotal Role of Women in Informal Care. J Fam Issues. 2008;29:1348–78. https://doi.org/10.1177/0192513X08316115.
43. Agbalajobi DT. The role of African women in peace building and conflict resolution: The case of Burundi. Afr J Psychol Study Social Stud. 2009;2:1–16.
44. Krug E, Cieza A. Strengthening health systems to provide rehabilitation services. Bull World Health Organ. 2017;95:167–167. https://doi.org/10.2471/BLT.17.191809.
45. World Physiotherapy. Ethiopia: a profile of the profession in 2023 [16 March, 2024]. Available from: https://world.physio/membership/ethiopia
46. Stenberg G, Ahlgren C. A gender perspective on physiotherapy treatment in patients with neck and back pain. Adv Physiother. 2010;12:35–41. https://doi.org/10.3109/14038190903174270.
47. Footer CB, Tsegaye HS, Yitnagashaw TA, Mekonnen W, Shiferaw TD, Abera E, et al. Empowering the Physiotherapy Profession in Ethiopia through Leadership Development within the Doctoring Process. Front Public Health. 2017;5. https://doi.org/10.3389/fpubh.2017.00051.
48. Sayeed MSI, Oakman J, Stuckey R. Factors influencing access to and participation in rehabilitation for people with lower limb amputation in East, South, and Southeast Asian developing countries: the perspective of rehabilitation professionals – a qualitative study. Disabil Rehabil. 2023;46(10):2097–116.
49. De Silva V. Its name is *Awetasc* ': devices and the everyday life of people with physical disability. Ethiopia Afr. 2022;92:522–39. https://doi.org/10.1017/S0001972022000420.
50. Wen P-S, Randolph MG, Elbaum L, De la Rosa M. Gender Differences in Psychosocial and Physical Outcomes in Haitian Amputees. Am J Occup Therapy. 2018;72:p72032050901–8. https://doi.org/10.5014/ajot.2018.022962.
51. Consolidating Logistics for Assistive Technology Supply and Provision (CLASP). Overcoming gender-related barriers to wheelchair services. Momentum Wheels for Humanity; 2017.
52. Hisayo K, Mina C, Mojtahedi. Intersection of disability and gender: multi-layered experiences of Ethiopian women with disabilities. Dev Pract. 2015;25(4):563–73. https://doi.org/10.1080/09614524.2015.1031085.
53. Baynesagn AH, Tolla EH. Poor Mothers and Begging: How Impoverished Ethiopian Women Support Their Children in the Absence of a Strong State Welfare System. Sage Open. 2022;12:215824402210917. https://doi.org/10.1177/21582440221091725.
54. Mckechnie PS, John A. Anxiety and depression following traumatic limb amputation: A systematic review. Injury. 2014;45:1859–66. https://doi.org/10.1016/j.injury.2014.09.015.
55. Şimsek N, Öztürk GK, Nahya ZN. The Mental Health of Individuals With Post-Traumatic Lower Limb Amputation: A Qualitative Study. J Patient Exp. 2020;7:1665–70. https://doi.org/10.1177/2374373520932451.
56. American Psychological Association, APA Task Force on Guidelines for Assessment and Intervention with Persons with Disabilities. (2022). Guidelines for Assessment and Intervention with Persons with Disabilities. Retrieved from https://www.apa.org/about/policy/guidelines-assessment-intervention-disabilities.pdf
57. Tsengu DV, Brodtkorb S, Almnes T. CBR and Economic Empowerment of Persons with Disabilities: [16 March, 2024]. Available from: https://asksource.info/cbr-book/cbraspart_04.pdf
58. Ott J, Champagne SN, Bachani AM, Morgan R. Scoping 'sex' and 'gender' in rehabilitation: (mis)representations and effects. Int J Equity Health. 2022;21:179. https://doi.org/10.1186/s12939-022-01787-1.

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